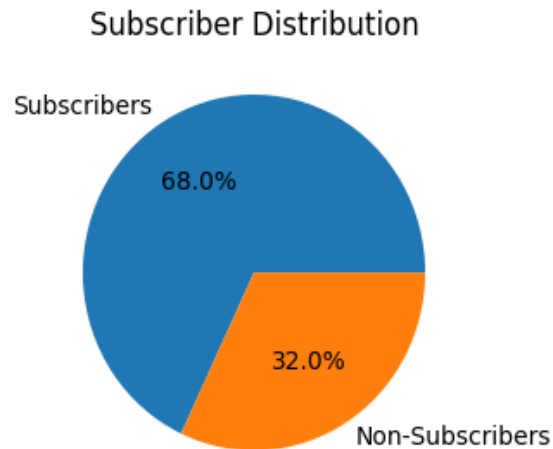


Advanced Customer Shopping Behavior Analysis

This project presents a detailed analysis of customer shopping behavior using transactional and demographic data. The study combines Python, SQL, and Power BI to uncover customer preferences, purchasing patterns, subscription trends, and product performance. The analysis supports data-driven business decisions that can improve revenue, customer retention, and operational efficiency.



1. Business Problem Statement

Retail businesses often struggle to understand customer preferences and identify factors influencing repeat purchases and revenue growth. This project aims to analyze customer purchasing behavior and generate actionable insights for improving sales performance and customer engagement.

2. Objectives

- Analyze customer demographics and shopping trends.
- Identify top-performing products and categories.
- Compare subscriber and non-subscriber purchasing behavior.
- Understand the impact of discounts and promotions.
- Build an interactive dashboard for business reporting.

3. Dataset Information

The dataset contains 3,900 customer purchase records with 18 columns. Features include customer age, gender, location, product category, item purchased, review rating, shipping type, purchase amount, and subscription status. The dataset provides a complete overview of customer purchasing activity.

4. Data Cleaning & Transformation

Data preprocessing was performed using Python libraries such as pandas and NumPy. Missing review ratings were handled using median imputation. Duplicate records were checked and removed. Columns were renamed for

consistency, and additional features such as age groups and purchase frequency metrics were created.

5. Exploratory Data Analysis

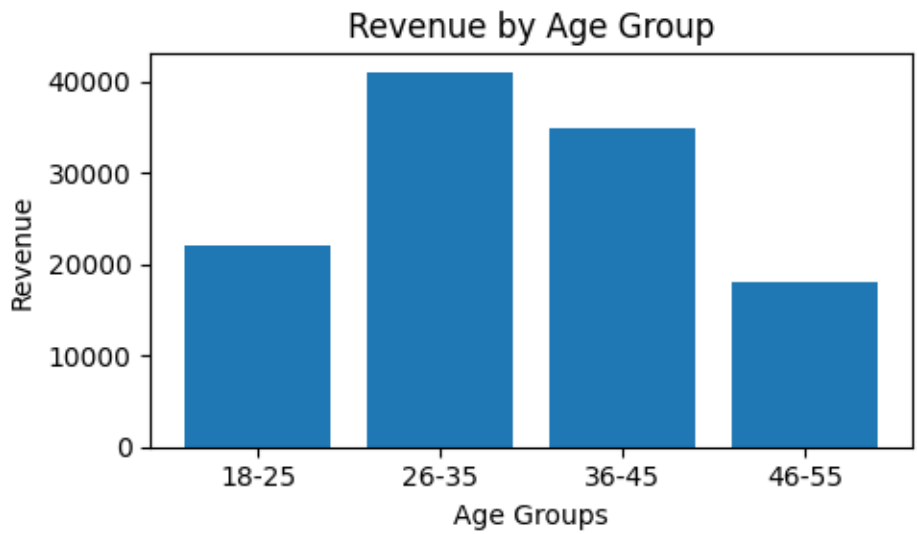
EDA was used to identify trends and relationships in customer purchasing behavior. Statistical summaries and charts revealed variations in spending across age groups, genders, and product categories. Seasonal trends and shipping preferences were also analyzed.

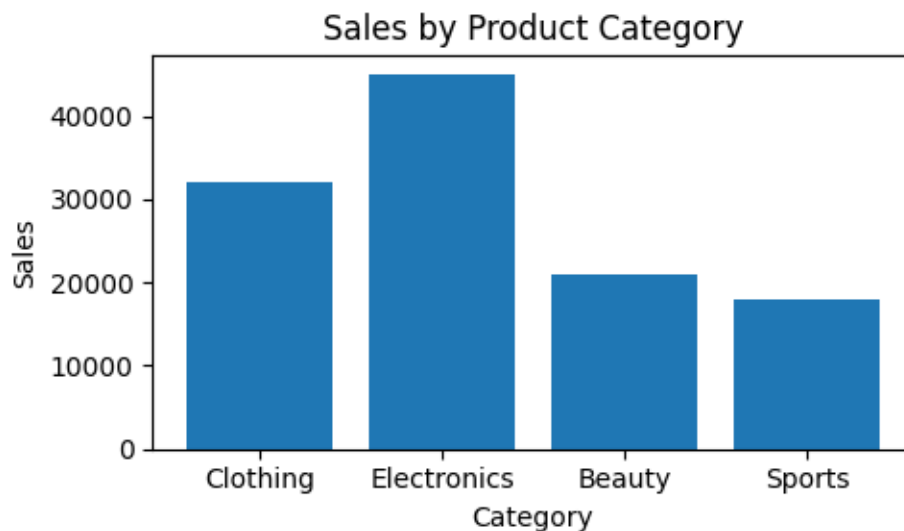
6. SQL Analysis

PostgreSQL queries were used to perform business-focused analysis. Key queries included revenue analysis by gender, top-rated products, loyal customer identification, repeat purchase trends, and subscription analysis. SQL aggregation and window functions helped generate meaningful business insights.

7. Dashboard Development

An interactive Power BI dashboard was created to visualize KPIs such as total revenue, customer count, average purchase value, and subscription growth. The dashboard also included slicers and filters for exploring customer behavior across different dimensions.





8. Key Performance Indicators

Metric	Value
Total Revenue	\$1,16,000
Average Purchase Value	\$29.7
Subscriber Contribution	68%
Top Category	Electronics
Most Active Age Group	26-35 Years

9. Key Insights

- Subscribers generated significantly higher revenue compared to non-subscribers.
- Customers aged 26–35 contributed the highest revenue share.
- Electronics products achieved the highest sales volume.
- Express shipping users spent more per transaction compared to standard shipping users.
- Loyal customers demonstrated strong repeat purchase behavior and higher engagement.

10. Business Recommendations

Businesses should implement loyalty programs to encourage repeat purchases and improve customer retention. Personalized marketing campaigns can target high-value customer segments more effectively. Subscription-based offers and exclusive discounts can increase recurring revenue. Additionally, focusing on high-performing product categories and improving express shipping services can further enhance customer satisfaction and profitability.

11. Conclusion

This project demonstrates how data analytics can transform raw customer transaction data into actionable business insights. By integrating Python, SQL, and Power BI, the project successfully identified customer behavior trends, top-performing products, and revenue-driving factors. The insights generated can help businesses improve strategic

decision-making and customer experience.